

“AI-Based Smart Microgrid Energy Management and Load Balancing Platform”

CO-1 ASSESSMENT TOOL-2

CASE STUDY ANALYSIS REPORT

Submitted in the partial fulfilment for the course of

CSA1017 - SOFTWARE ENGINEERING

to the award of the degree of

BACHELOR OF ENGINEERING

IN

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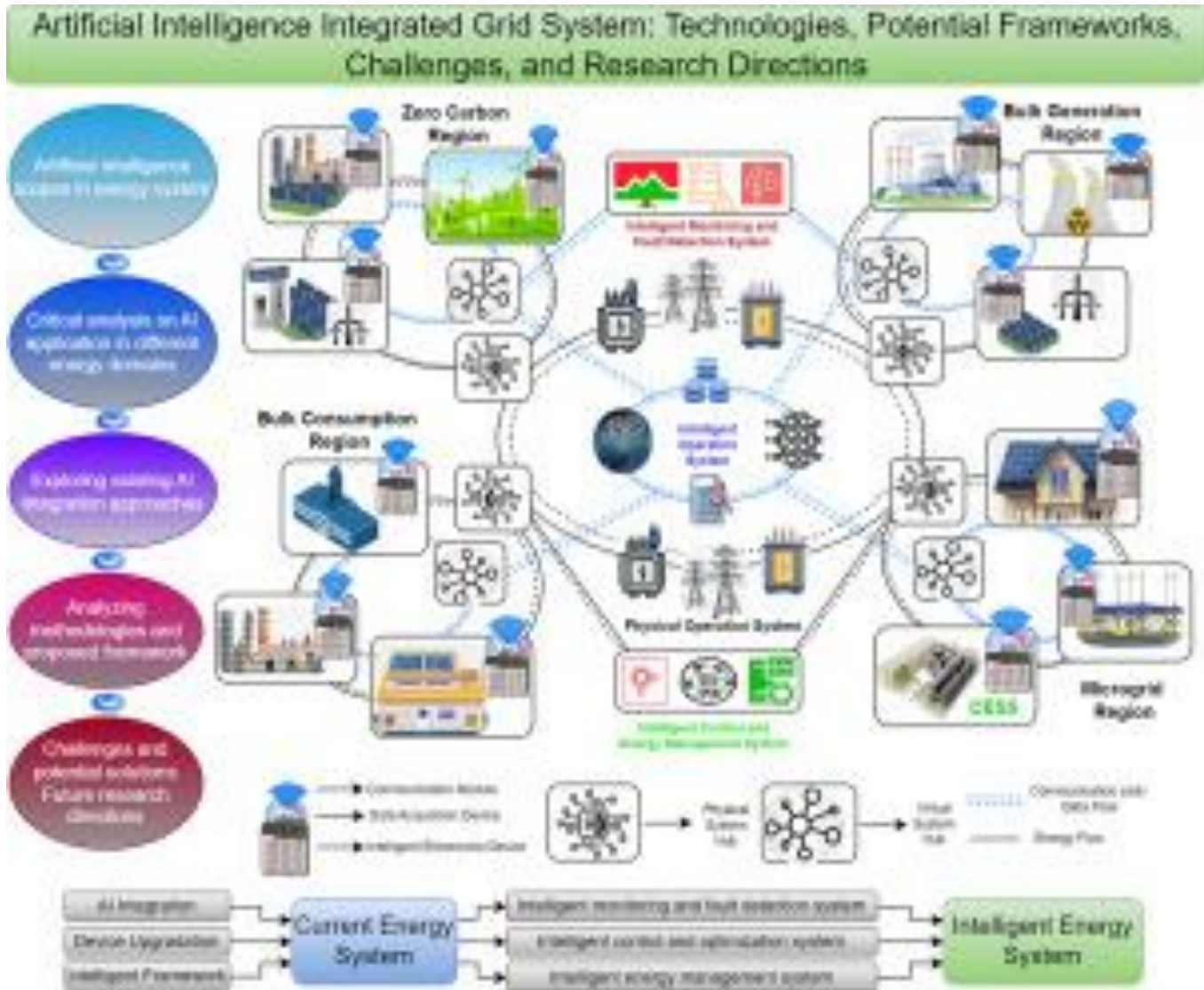
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1.Introduction

The AI-Based Smart Microgrid Energy Management and Load Balancing Platform is a smart energy management system that uses Artificial Intelligence (AI), IoT, and cloud technology to monitor, predict, and optimize electricity generation, storage, and consumption. It helps balance electrical loads, improve energy efficiency, reduce operational costs, and maximize the use of renewable energy sources. The project follows the Agile Software Development approach to deliver reliable and sustainable energy management solutions.



2. Project Vision and Stakeholders

2.1 Project Vision

Develop an **AI-Based Smart Microgrid Energy Management and Load Balancing Platform** that intelligently monitors, predicts, and optimizes electricity generation, storage, and consumption. The platform will leverage Artificial Intelligence (AI), Internet of Things

(IoT), Smart Meters, and Cloud Computing to improve energy efficiency, reduce operational costs, enhance grid reliability, and maximize renewable energy utilization.

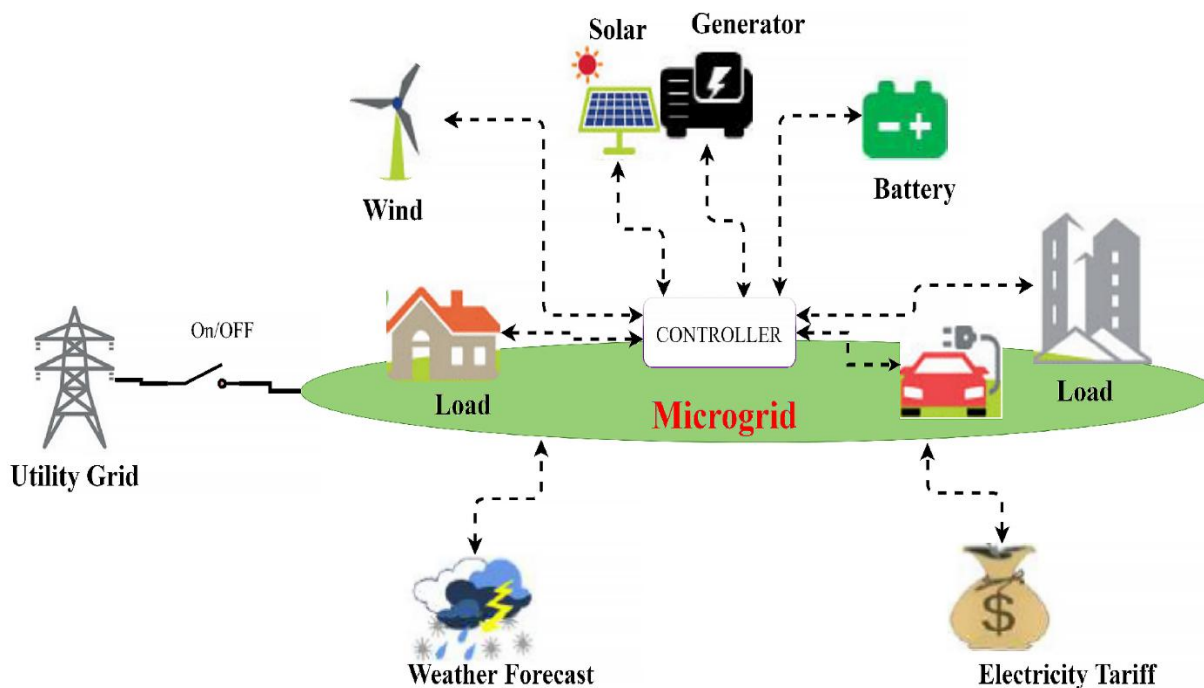
2.2 Project Objectives

- Monitor real-time energy generation and consumption.
- Forecast future electricity demand using AI.
- Optimize battery storage charging and discharging.
- Automatically balance electrical loads.
- Integrate renewable energy sources such as solar and wind.
- Generate energy usage and performance reports.
- Reduce operational costs.
- Improve energy reliability and sustainability.

2.3 Stakeholders

Stakeholder	Role
Microgrid Administrator	Monitors and manages the complete microgrid system
Energy Operator	Controls energy distribution and monitors performance
Residential/Industrial Consumers	View electricity usage and receive energy recommendations
Renewable Energy Providers	Supply solar/wind energy to the microgrid
Utility Company	Coordinates grid connectivity and backup supply
Maintenance Engineers	Maintain sensors, batteries, and equipment

Stakeholder	Role
Government & Regulatory Authorities	Ensure compliance with energy regulations
System Administrator	Manages users, security, and system configuration



2.4 Epics and User Stories

Epic 1 – User Authentication & Access Management

User Story 1

As a System Administrator, I want secure login authentication so that only authorized users can access the platform.

User Story 2

As a System Administrator, I want role-based access control so that different users have appropriate permissions.

Epic 2 – Real-Time Energy Monitoring

User Story 3

As an Energy Operator, I want to monitor electricity generation and consumption in real time so that I can identify abnormal energy usage.

User Story 4

As a Consumer, I want to view my electricity consumption dashboard so that I can understand my energy usage.

Epic 3 – AI Demand Forecasting

User Story 5

As an Energy Operator, I want AI to forecast electricity demand so that I can prepare sufficient power generation.

User Story 6

As a Microgrid Administrator, I want predictive analytics reports so that I can plan future energy requirements.

Epic 4 – Battery Storage Optimization

User Story 7

As an Energy Operator, I want the system to optimize battery charging and discharging so that stored energy is efficiently utilized.

Epic 5 – Automatic Load Balancing

User Story 8

As a Microgrid Administrator, I want automatic load balancing so that electrical overloads are prevented.

User Story 9

As a Consumer, I want uninterrupted electricity during peak hours so that my operations continue without interruption.

Epic 6 – Renewable Energy Integration

User Story 10

As a Renewable Energy Provider, I want the platform to integrate solar and wind energy so that renewable resources are efficiently utilized.

Epic 7 – Reports & Dashboard

User Story 11

As an Administrator, I want energy performance reports so that I can analyze operational efficiency.

User Story 12

As a Consumer, I want monthly energy usage reports so that I can reduce electricity expenses.

3. Acceptance Criteria (Given–When–Then)

User Story 1

Given the administrator has valid credentials

When they log into the system

Then access is granted to the dashboard.

User Story 3

Given energy sensors are active

When electricity data is received

Then the dashboard updates generation and consumption in real time.

User Story 5

Given historical electricity data is available

When AI forecasting runs

Then predicted electricity demand is displayed with accuracy.

User Story 7

Given battery charge information is available

When battery optimization executes

Then charging/discharging is automatically optimize

User Story 8

Given electrical load exceeds the threshold

When overload is detected

Then loads are automatically redistributed.

User Story 10

Given renewable energy is generated

When solar or wind energy becomes available

Then the system integrates renewable energy into the microgrid.

User Story 11

Given operational data exists

When the administrator requests reports

Then the system generates downloadable energy performance reports.

4. Product Backlog Prioritization

The product backlog is prioritized based on business value, customer needs, and project dependencies. The highest-priority items include User Login & Authentication, Real-Time Energy Monitoring, AI Demand Forecasting, Battery Storage Optimization, and Automatic

Load Balancing, as these are the core functionalities of the platform. These features depend on components such as IoT sensors, historical data, battery sensors, and the monitoring module.

The medium-priority items are Renewable Energy Integration, Performance Dashboard, and Energy Reports, which enhance the platform by supporting renewable energy management, providing data visualization, and generating performance reports. These features depend on solar and wind energy sources, the monitoring module, and the database.

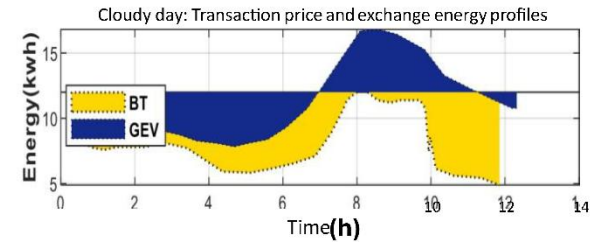
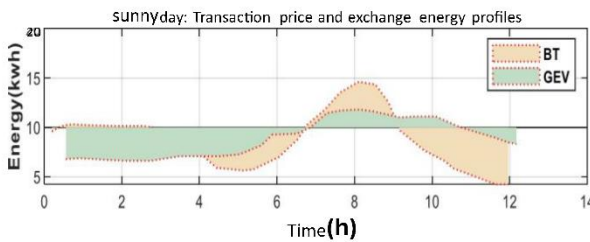
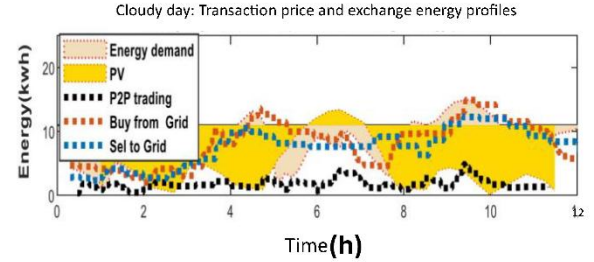
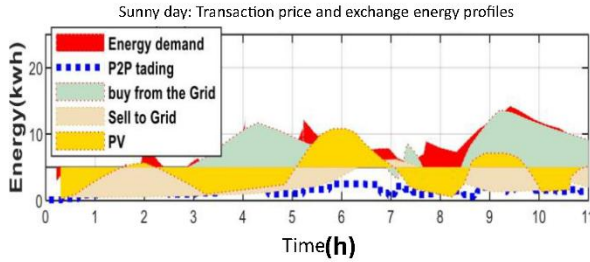
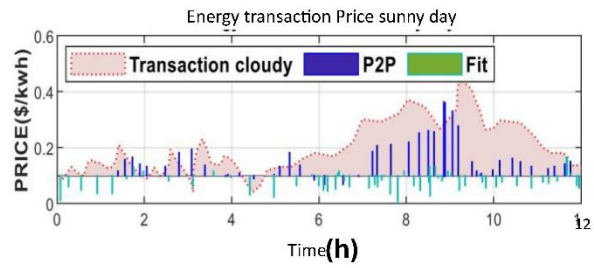
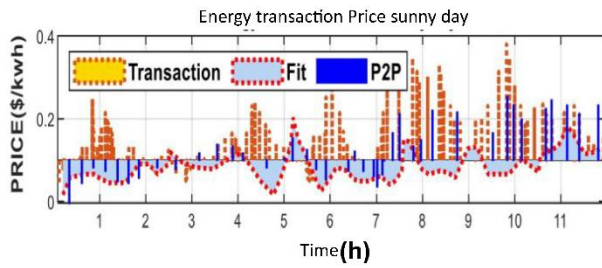
The low-priority item is Consumer Notifications, which provides alerts and updates to users. This feature depends on the successful implementation of the dashboard and is scheduled after the core functionalities are completed.

5. Sprint Backlog (Sprint 1)

5.1 Sprint Duration

2 Weeks

User Story	Development Tasks
User Authentication	Design login page, database, authentication API, testing
Energy Monitoring	Integrate IoT sensors, display dashboard, data validation
Dashboard	Develop UI, charts, live updates
AI Forecast Preparation	Collect historical energy data, preprocess dataset
Database	Create user, sensor, battery, and report tables



6. Story Point Estimation (Fibonacci)

User Story	Story Points	Justification
User Authentication	3	Standard login and security implementation
Energy Monitoring	8	IoT integration and real-time updates are complex
Dashboard	5	Multiple visualizations and user interface development
AI Demand Forecasting	13	AI model development and prediction require significant effort

User Story	Story Points	Justification
Battery Optimization	8	Requires optimization algorithms and battery monitoring
Load Balancing	13	High complexity involving automation and decision-making
Renewable Energy Integration	8	External device and renewable source integration
Reports	5	Report generation and export functionality

Estimation Technique

Fibonacci Sequence (1, 2, 3, 5, 8, 13)

Justification

- **3 Points:** Small implementation with minimal complexity.
- **5 Points:** Moderate development with UI and backend integration.
- **8 Points:** Complex functionality requiring IoT integration.
- **13 Points:** Highly complex AI algorithms and automation.

7. Sprint Goal and Expected Deliverables

Sprint Goal

Develop the foundational features of the **AI-Based Smart Microgrid Energy Management and Load Balancing Platform**, including secure user authentication, real-time energy monitoring, dashboard visualization, database setup, and preparation for AI-based demand forecasting.

Expected Sprint Deliverables

- Secure Login & User Authentication
- Role-Based User Management
- Real-Time Energy Monitoring Dashboard
- Database for Users, Sensors, and Energy Data

- Initial AI Demand Forecasting Dataset Integration
- Interactive Dashboard with Live Charts
- Basic Performance Report Module

Sprint Summary

Sprint	Sprint 1
Duration	2 Weeks
Sprint Goal	Build the core infrastructure and real-time monitoring features
Total User Stories	5
Estimated Story Points	34 Story Points
Expected Outcome	Functional prototype with secure login, energy monitoring dashboard, database, and AI forecasting preparation

7.Conclusion

The AI-Based Smart Microgrid Energy Management and Load Balancing Platform applies Agile software engineering practices by organizing requirements into epics, user stories, prioritized backlogs, and sprint tasks. The first sprint focuses on delivering the core system infrastructure, enabling secure access, real-time monitoring, and the groundwork for AI-powered demand forecasting. This iterative Agile approach supports continuous improvement, efficient energy management, increased renewable energy utilization, reduced operational costs, and enhanced microgrid reliability.